

IISER Mohali -PHY102-Electromagnetism

Assignment-1

Due date: 24th January 2025, before 3pm, upload in moodle

Q1. What is free charge and bound charge?, give an example for each case. Draw/give a design that can be used to separate induced charges (positive and negative type) into two separate objects respectively. In an another situation, assume, you have only one metal sphere- how will you make a single metal sphere a positive induced charge source?

Q2. What is Coulomb's law? Define Electric Field $[E(r)]$ at r in 3D space due to charge $q(r')$ at r' . A charge Q is divided into two parts such that they repel each other with a maximum force when placed at a certain distance apart. Find the distribution of charge.

Q3. Two identical metallic spheres, having unequal opposite charges are placed at a distance of 0.60 metre apart in air. After bringing them in contact with each other, they are again placed at the same distance apart. Now the force of repulsion between them is 0.366 N. Calculate the final charge on each of them.

Q4. Show that the gravitational force is negligible in comparison to electric force in hydrogen atom, in which the electron and proton are about 5.3×10^{-10} metre apart. mass of electron = $9.1 \times 10^{-31} \text{ Kg}$, mass of proton = $1.6 \times 10^{-27} \text{ Kg}$, $r = 5.3 \times 10^{-10} \text{ m}$ and $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{Kg}^2$

Q5. Compute the electric field intensity at a point 60 cm away in vacuum from an electric charge of $4 \times 10^{-9} \text{ C}$. An electron covers a distance of 60 mm when accelerated from rest by an electric field of intensity $2 \times 10^4 \text{ N/C}$. Calculate the time of travel. (The mass of electron = $9 \times 10^{-31} \text{ Kg}$, Charge on electron = $1.6 \times 10^{-19} \text{ C}$ respectively)

Q6. Two point charges of $5 \times 10^{-19} \text{ C}$ and $20 \times 10^{-19} \text{ C}$ are separated by a distance 1 meter. At which point on the line joining them, the electric field is zero? If a charge $12 \times 10^{-19} \text{ C}$ is placed at this point then what will be the force acting on it?

Q7. How much work is done in bringing a charge of $5.0 \times 10^{-6} \text{ C}$ from one point to another, if the potential difference between the two points is 5 volt?

Q8. What is the intensity of electric field due to a helium nucleus at a distance of 1 A^0 from the nucleus? The electric field intensity at a point on the line joining two point charges is zero. What conclusion can you draw about the charges?

Q9. Two point charges $+6 \text{ } \mu\text{C}$ and $-2 \text{ } \mu\text{C}$ are 0.8 m apart. Locate the point at which the electric potential is zero.

If electric field intensity is zero at a point, then Will the electric potential be necessarily zero at that point?

Q10. One coulomb charge is kept at the centre of a cube of side 5 cm. Find out the electric flux coming out of any one face of the cube.

Two electric lines of force never intersect each other. Why?

Q10. Using Gauss' theorem, estimate/derive the electric field intensity due to a charged spherical shell at a point outside the shell.